8 LAND CONTAMINATION IMPACT

8.1 Introduction

8.1.1 Contaminated land refers to the land which has been polluted by hazardous substances as a result of industrial or commercial operations carried out on site over a number of years. These contaminants, if present, may pose hazardous risks or cause adverse effects to the land users and the nearby environment. The implications of land contamination associated with the Project were assessed in accordance with Section 3.4.9 of the EIA Study Brief (No. ESB-291/2015).

8.2 Environmental Guidelines, Policies, Plans, Standards and Criteria

8.2.1 Relevant environmental guidelines and standards on land contamination aspect include the following:

- Section 3 (Potential Contaminated Land Issues) of Annex 19 of the 'Guidelines for Assessment of Impact on Sites of Cultural Heritage and Other Impacts' from the Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM).

- The Guidance Note for Contaminated Land Assessment and Remediation (Guidance Note) sets out the requirements for proper assessment and management of potentially contaminated sites such as oil installations (e.g. oil depots, petrol filling stations), gas works, power plants, shipyards/boatyards, chemical manufacturing / processing plants, steel mills/metal workshops, car repairing/dismantling workshops and scrap yards.

- The Practice Guide for Investigation and Remediation of Contaminated Land (Practice Guide) outlines typical investigation methods and remediation strategies for the range of potential contaminants typically encountered in Hong Kong.

- The Guidance Manual for Use of Risk-Based Remediation Goals for Contaminated Land Management (Guidance Manual) presents the risk based approach for contamination assessment and instructions for comparison of soil and groundwater data with Risk-Based Remediation Goals (RBRGs) for 54 chemicals of concern commonly found in Hong Kong. The RBRGs were derived to suit Hong Kong conditions and were designed to protect the health of people who could potentially be exposed to land impacted by chemicals under four broad post-restoration land use categories. RBRGs also serve as the remediation targets if remediation is necessary.

8.3 Assessment Approach and Methodology

8.3.1 The following tasks have been undertaken to identify any potentially contaminated areas for the land contamination assessment:

- Desktop review of site history; and

- Site surveys to identify the potentially contaminated sites.

8.3.2 For the desktop review, the following information has been collated and reviewed:

- Hong Kong Ordinance Survey Maps;

- Relevant aerial photographs;

- Relevant Outline Zoning Plans (OZPs);
- Hong Kong Geological Survey Maps (GSMs);
- Records of dangerous goods, chemical wastes and chemical spillage/leakage incident from Fire Services Department (FSD) and EPD; and
- "Planning and Development Study on North West New Territories – Final Environmental Impact Assessment", Territory Development Department and Planning Department, 2001.

8.3.3 Findings of the desktop review and site surveys, as well as the proposed soil and groundwater sampling and testing strategy are presented in the Contamination Assessment Plan (CAP) enclosed in **Appendix 8.1**.

8.3.4 Due to the physical constraints and accessibility issues of the identified potentially contaminated sites, undertaking site investigation (SI) works at this EIA stage is not feasible. Information as required in S.3.4.9.1(vi) of the EIA Study Brief was subsequently presented in the CAP.

8.3.5 Further works, including site re-appraisal and SI works (refer to **Section 8.9**), are recommended to be conducted in the later stage of this Project. Supplementary CAP(s), incorporation of findings of the site re-appraisal and the updated sampling and testing strategy will be submitted to EPD for approval prior to conducting any SI works.

8.3.6 Following the completion of SI works, Contamination Assessment Report(s) (CAR(s)) and, if necessary, Remediation Action Plan(s) (RAP(s)) will be submitted to EPD for approval to identify the presence, nature and extent of contamination as well as the proposed remedial strategy for the identified contaminated soil and/or groundwater. Any contaminated soil and/or groundwater should be treated according to EPD's approved RAP(s) and Remediation Report(s) (RR(s)) will be submitted to EPD for agreement after completion of the remediation works.

8.4 Description of the Project and Assessment Area

8.4.1 The Project Area of the Hung Shui Kiu New Development Area (HSK NDA) in the EIA Study Brief (No. ESB-291/2015) (referred to hereafter as Project Area), which covers an area of about 714 hectares, is bounded by Tin Ying Road and Kiu Hung Road to the east; Castle Peak Road to the south; Lau Fau Shan Road and Deep Bay Road to the north; and Kong Sham Western Highway (KSWH) to the west. In order to support the Project, a fresh water and flushing water service reservoir (hereinafter refer to as Tan Kwai Tsuen Reservoir) has been proposed. The Tan Kwai Tsuen Service Reservoir (depicted as 'Reservoir A' in the CAP), which will be constructed on the southeastern side of the Project Area, will be located on a hillside outside the Project Area. The footprint of the Project Area and Tan Kwai Tsuen Service Reservoir is shown in **Figure 1.1**.

8.4.2 In addition to the above and as agreed by EPD, as there will not be any development and/or redevelopment works in the existing recognised villages within the Project Area, the land contamination assessment will exclude these recognised village areas. The assessment area for land contamination assessment (i.e. Tan Kwai Tsuen Service Reservoir and Project Area excluding the recognised village areas) is shown in **Figure 8.1**.

8.5 Baseline Conditions

Review of Historical Aerial Photos

8.5.1 A comprehensive review of aerial photographs has been undertaken under this Project using a series of five aerial photographs covering the period 1978 – 2013. The aim of the review was to evaluate the likelihood of potential contamination associated with past land uses within the assessment area. The development history of the assessment area is summarised in **Table 8.1** and the historical aerial photographs are provided in Appendix A1 of the CAP given in **Appendix 8.1**.

Table 8.1 Summary of Historical Land Uses for the Assessment Area

Year	Historical Land Use
1978	• The majority of the Project Area comprised natural terrain, rural residential areas and agricultural land. • Castle Peak Road was identified. • Tan Kwai Tsuen Service Reservoir location comprised natural terrain. No industrial activities were observed.
1985	• No significant change in land use was observed in the Project Area in comparison with land use from the 1978 aerial photographs. • Tan Kwai Tsuen Service Reservoir location comprised mainly natural terrain and an access road. No industrial activities were observed.
1995	• Predominantly agricultural lands in the northern, eastern and central portions of the Project Area were gradually replaced by industrial activities such as open area storage and warehousing. • No significant change in land use was observed at the reservoir location in comparison with land use from the 1985 aerial photograph. No industrial activities were identified.
2010	• Predominantly agricultural lands in the north-western and north-eastern portions of the Project Area were replaced by industrial activities. • KSWH and MTR West Rail Line were identified in the western and southern portions of the area respectively. • No significant change in land use was observed at the reservoir location in comparison with land use from the 1995 aerial photograph. No industrial activities were identified.
2013	• No significant change in land use was observed in the Project Area in comparison with 2010 aerial photographs. • No significant change in land use was observed at the reservoir location in comparison with land use from the 2010 aerial photograph. No industrial activities were identified.

Site Geology

8.5.2 Ground investigation (GI) works have been carried out over the assessment area for various projects and developments and a total of 3,001 nos. of borehole records were retrieved from the Geotechnical Engineering Office of CEDD and reviewed. Based on the existing GI records, fill deposits beneath developed areas within the assessment area vary in thickness from 1 m and 5 m.

8.5.3 The majority of the assessment area is covered with a layer of terraced alluvium except for some areas in the northwest and southwest where outcrops of *in-situ* materials are exposed. The alluvium underlying the fill, marine deposits or pond deposits are variable in terms of their composition and typically described as soft to stiff sandy/silty clay and clayey silt, and loose to very dense sand and gravel with cobbles. The thickness of alluvium encountered within the assessment area typically varies from 0.3 m to over 24 m.

8.5.4 A layer of *in-situ* soil, predominately saprolite and in places Grade IV rock is located between the alluvium and rockhead. The predominant unit of the saprolite is Grade V materials derived from weathered rocks. The thickness of saprolite varies from less than 1 m to over 115 m.

8.5.5 Groundwater level monitoring records from standpipes and piezometers in the Geotechnical Assessment and Site Investigation Report in the assessment area were also reviewed. Records show that the groundwater levels are generally shallow at about 2 m below the ground surface in flat land. No groundwater monitoring records were reviewed for natural terrain within the assessment area.

Site Appraisal

8.5.6 Helicopter reconnaissance of the assessment area was undertaken in September 2014, with extensive video and photographs taken that allowed identification of current land uses. This survey data was supplemented with site visits undertaken in December 2014 to confirm current land uses within the assessment area.

8.5.7 Based on the results of desktop review, helicopter reconnaissance and site visits, a total of 522 sites have been identified for further study. However, as most of the sites are occupied by private owners and inaccessible for site walkover, only 15 sites were accessible to conduct thorough site walkover. For the remaining sites, peripheral observations were conducted during the site visits. A summary of the 522 surveyed sites is given in **Table 8.2**.

Table 8.2 Summary of Surveyed Sites

Descriptions	No. of Sites
Surveyed Sites *Accessible* for Site Walkover	
Potentially Contaminated Sites	3
Non-Contaminated Sites	12
Sub-total	15
Surveyed Sites *Inaccessible* for Site Walkover	
Potentially Contaminated Sites	477
Non-Contaminated Sites	30
Sub-total	507
Total No. of Sites	522

8.5.8 Additionally, no industrial activities were found at the Tan Kwai Tsuen Service Reservoir location based on the review of historical aerial photographs (refers to Appendix A1 of the CAP given in **Appendix 8.1**). Furthermore, the findings of the site survey indicated that the current land use at reservoir location is natural terrain. No land contamination issue is therefore anticipated within location of the reservoir.

8.5.9 Based on the site appraisal, the following land uses with the potential to cause land contamination issues were identified:

- Open area storage;
- Container storage;
- Warehouses;
- Construction equipment storage / equipment depot;
- Factories;
- Recycling facilities;
- Petrol filling stations;
- Vehicle / repair maintenance workshops; and
- Concrete batching plants.

8.5.10 With reference to the EPD's *Practice Guide*, the key Chemicals of Concern (COCs) and remediation methods associated with the identified potentially contaminating land uses are identified and summarised in **Table 8.3**.

Table 8.3 Potential Chemicals of Concern (COCs) and Possible Remediation Methods for the Identified Potentially Contaminating Land Uses

Identified Potentially Contaminating Land Uses	Potentially Contaminating Activities	COCs(1)	Possible Remediation Methods(2)
Open area storage, container storage and warehouse	Loading, unloading and storage of goods, fuel storage and transfer, maintenance of equipment and vehicles.	Metals (full list), PCRs, VOCs and SVOCs.	Biopiling, Solidification / Stabilisation, Permeable Reactive Barriers
Construction equipment storage / equipment depot and vehicle / repair maintenance workshops	Release of oils, fuels and lubricants from vehicles, vehicle and equipment maintenance and refuelling. Use of chemicals and solvents in maintenance activities. Motor vehicle painting and storage and disposal of wastes.	Metals (e.g. chromium, copper, lead, manganese, nickel, zinc), PCRs, VOCs (e.g. acetone, BTEX, MTBE, and trichloroethene) and PAHs.	Biopiling, Solidification / Stabilisation, Permeable Reactive Barriers
Factories	Release of oils, fuels and lubricants from mechanical machinery, equipment maintenance and refuelling. Use of chemicals and solvents in maintenance activities.	Metals (full list), PCRs, VOCs and SVOCs.	Biopiling, Solidification / Stabilisation, Permeable Reactive Barriers
Recycling facilities	Storage and processing of waste materials, storage and transfer of chemicals and fuels. Storage and disposal of wastes.	Metals (full list), PCRs, VOCs, SVOCs and PCBs.	Biopiling, Solidification / Stabilisation, Permeable Reactive Barriers
Petrol filling stations	Leaks from pipework, tanks and offset fill pipes. Spills during customer refuelling, filling underground storage tanks and over filling of portable containers.	Metals (e.g. lead), PCRs, VOCs (e.g. BTEX and MTBE), and PAHs.	Biopiling, Solidification / Stabilisation, Permeable Reactive Barriers
Concrete batching plants	Storage and transfer of residues from physical conversion of earthen materials by sorting, mixing, and grinding.	VOCs (e.g. BTEX), SVOCs and PCRs.	Biopiling, Solidification / Stabilisation, Permeable Reactive Barriers

Note:
(1) - Metals include antimony, arsenic, barium, cadmium, chromium III, chromium VI, cobalt, copper, lead, manganese, mercury, molybdenum, nickel, tin and zinc.
- Petroleum carbon ranges (PCRs) include C6 – C8, C9 – C16 and C17 – C35.
- Volatile organic chemicals (VOCs) include BTEX (benzene, toluene, ethylbenzene, and total xylenes), MTBE (methyl tert-butyl ether), acetone, bromodichloromethane, 2-butanone, chloroform, methylene - chloride, styrene, tetrachloroethene, and trichloroethene.
- Semi-volatile organic chemicals (SVOCs) include polyaromatic hydrocarbons (PAHs) (acenaphthene, acenaphthylene, anthracene, benzo(a)anthracene, benzo(a)pyrene, benzo(b)fluoranthene, benzo(g,h,i)perylene, benzo(k)fluoranthene, chrysene, dibenzo(a,h)anthracene, fluoranthene, fluorene, indeno(1,2,3-cd)pyrene, naphthalene, phenanthrene and pyrene), bis-(2-ethylhexyl)phthalate, hexachlorobenzene, and phenol.
- PCBs - Polychlorinated biphenyls.
(2) Details of possible remediation method(s) for the COCs are listed in Section 5 of the CAP given in **Appendix 8.1**.

8.6 Identification of Potentially Contaminated Sites

Potentially Contaminated Sites Accessible for Site Walkover

8.6.1 Amongst all the sites which were accessible for site walkover, A78 (site office with maintenance workshop), E63a (open area storage of recycling materials) and E63b (bus parking area with petrol filling station) have been identified as potentially contaminated and require further site investigation. The locations of A78, E63a and E63b are shown in **Figures 8.2** to **8.5**. Details of A78, E63a and E63b are summarised in **Table 8.4** and the corresponding sampling and testing strategies are presented in **Section 8.7**.

Potentially Contaminated Sites Inaccessible for Site Walkover

8.6.2 Based on the site appraisal, a total of 507 sites were inaccessible for site walkover, in which 477 sites are considered to be potentially contaminated and further site investigations are required.

8.6.3 Among the potentially contaminated sites, 253 of them are currently used as open storage area, container storage and warehouse. From the observations of helicopter reconnaissance and site visits, open storage area and container storage comprise a large portion of area for goods or container storage. Any contaminating activities such as vehicle / equipment maintenance area and the associated chemical handling/storage would occupy a relatively small portion of the site. The extent of contamination (if any) within these sites would thus greatly depend on the types of goods stored on-site. For example, inert materials (e.g. brick and wood) stored in the open storage area would not cause contamination and any contamination would be restricted only to their small-scale maintenance workshop or other small-scale contaminating activities.

8.6.4 For the aforementioned open area storage, container storage and warehouse sites, 90 sites are used as warehouse with no other historical potential contaminated land uses identified. For these 90 sites, the presence and degree of contamination would greatly depend on the types of goods stored within these warehouses. For example, warehouses that stored general household goods (e.g. furniture and toys) are unlikely to cause contamination to the underlying soil and groundwater. A site re-appraisal within these sites is therefore required at the later stage of the Project to: (i) assess the nature of these warehouses, site conditions and the historical usage, (ii) confirm the necessity for site investigation works and, if required, (iii) formulate the sampling and testing strategies.

8.6.5 Since the land contamination potential for the 90 ‘warehouse’ sites cannot be confirmed at this stage, the corresponding sampling and testing strategies would be proposed subject to the results of site re-appraisal after land resumption. For the remaining 387 sites, the sampling and testing strategies are discussed in **Section 8.7**.

8.6.6 It is also worth noting that, apart from the aforementioned open area storage, container storage and warehouse sites, the land uses of the remaining potentially contaminated sites identified are not large scale polluting installations / facilities, such as oil depot and power plant. In fact, more than 70% of the remaining sites are less than 3,000 m^2 in area which is considered relatively small in scale.

Table 8.4 Summary of Potentially Contaminated Sites Accessible for Site Walkover within the Assessment Area

Site ID	Current Land Use / Activities	Approximate Area (m^2)	Site Observations / Information from FSD and EPD	Historical Potentially Contaminating Activities	Concerned Area	Proposed Number of Boreholes[1][2]	COCs[3]	Possible Remediation Methods[4]	Future Land Use	Relevant RBRGs Land Use Scenario
A78	Site Office, Construction Material Storage and Maintenance Workshop	1,860	Site office and open area storage of inert construction materials were observed. A maintenance workshop (i.e. with approximate area of 50 m^2) was observed in the eastern side of the site. The site was concrete paved. The possible contaminating activities may include release of oils and fuels and lubricants from equipment maintenance in the workshop. Use of chemicals and solvents in maintenance activities. Based on FSD/EPD information, there were no dangerous goods (DGs) / chemical wastes / incident / spillage records for the site.	No potentially contaminating activities were identified.	Maintenance Workshop (50 m^2)	2	Metals (e.g. chromium, copper, lead, manganese, nickel, zinc), PCRs, VOCs, (e.g. acetone, BTEX, MTBE, and trichloroethene) and SVOCs (e.g. PAHs)	Biopiling, Solidification / Stabilisation, Permeable Reactive Barriers	Institution and Community	Urban Residential
E63a	Open Area Storage of Recycling Materials	1,240	Temporary storage of recycling materials, such as unused rail tracks, wooden board, tyres, plastic pails and planks were observed. The site was unpaved and no burn pit was observed. According to the site representative, the site has been used as a temporary storage of recycling materials for more than 20 years. Considering the duration of its operation, further site investigation is recommended as a conservative approach. The possible contaminating activities may include storage of waste, fuel storage and transfer, maintenance of equipment and vehicles. Based on FSD/EPD information, there were no dangerous goods (DGs) / chemical wastes / incident / spillage records for the site.	*Open Area Storage* The possible contaminating activities may include loading, unloading and storage of goods, fuel storage and transfer, maintenance of equipment and vehicles.	Whole area of E63a (1,240 m^2)	9	Metals (full list), PCRs, VOCs, SVOCs and PCBs.	Biopiling, Solidification / Stabilisation, Permeable Reactive Barriers	Light Rail Facility, Urban Residential and Commercial	Urban Residential

Site ID	Current Land Use / Activities	Approximate Area (m^2)	Site Observations / Information from FSD and EPD	Historical Potentially Contaminating Activities	Concerned Area	Proposed Number of Boreholes[1][2]	COCs[3]	Possible Remediation Methods[4]	Future Land Use	Relevant RBRGs Land Use Scenario
E63b	Bus Parking Area with Petrol Filling Station and Diesel Storage Tank	5,140	Bus parking area and container offices were observed. A diesel fuel storage in a 10,000 L above ground tank sited on engineered pavement and within a steel containment bund was found in the southwestern part of the site. Protective drainage is located adjacent to the tank. One pump island and two diesel flowmeters are associated with the diesel tank. The area of the petrol filling station is around 40 m^2. No oil stain was observed. The possible contaminating activities may include leaks from pipework, tanks and offset fill pipes. Spills during customer refuelling, filling ground storage tanks and over filling of portable containers. Based on FSD/EPD information, there were no incident / spillage records for the site. However, there is an active chemical waste producer record and a registered 10,000 L aboveground diesel storage tank for the site.	No potentially contaminating activities were identified.	Petrol Filling Station (40 m^2)	2	Metals (e.g. Lead), PCRs, VOCs (e.g. BTEX and MTBE), and PAHs.	Biopiling, Solidification / Stabilisation, Permeable Reactive Barriers	Light Rail Facility, Urban Residential and Commercial	Urban Residential

Note:

(1) Details of the sampling strategy are listed in Table 4.1 of the CAP given in **Appendix 8.1**.

(2) The recommended sampling and testing strategy is only tentative and is subject to review after land resumption and site re-appraisal.

(3) - Metals include antimony, arsenic, barium, cadmium, chromium III, chromium VI, cobalt, copper, lead, manganese, mercury, molybdenum, nickel, tin and zinc.
- Petroleum carbon ranges (PCRs) include C6 – C8, C9 – C16 and C17 – C35.
- Volatile organic chemicals (VOCs) include BTEX (benzene, toluene, ethylbenzene, and total xylenes), MTBE (methyl tert-butyl ether), acetone, bromodichloromethane, 2-butanone, chloroform, methylene chloride, styrene, tetrachloroethene, and trichloroethene.
- Semi-volatile organic chemicals (SVOCs) include polyaromatic hydrocarbons (PAHs) (acenaphthene, acenaphthylene, anthracene, benzo(a)anthracene, benzo(a)pyrene, benzo(b)fluoranthene, benzo(g,h,i)perylene, benzo(k)fluoranthene, chrysene, dibenzo(a,h)anthracene, fluoranthene, fluorene, indeno(1,2,3-cd)pyrene, naphthalene, phenanthrene and pyrene), bis-(2-ethylhexyl)phthalate, hexachlorobenzene, and phenol.
- PCBs - Polychlorinated biphenyls.

(4) Details of possible remediation method(s) for the COCs are listed in Section 5 of this CAP given in **Appendix 8.1**.

8.7 Site Investigation Plan

Potentially Contaminated Sites Accessible for Site Walkover

8.7.1 Based on **Section 8.6,** three potentially contaminated sites (i.e. A78, E63a and E63b), accessible for site walkover, were identified. Site investigation (SI) is recommended to be carried out to determine the presence, nature and quantities of soil/groundwater contamination within the sites. The sampling and testing plan for each site has been determined based on the recommendations given in Section 2.4.1 of EPD's *Practice Guide* and summarised in **Table 8.5**. The proposed locations of the boreholes are shown in **Figures 8.6** to **8.8**. Details of the proposed SI works are discussed in CAP given in **Appendix 8.1**.

8.7.2 Permissions for conducting SI works have been verbally discussed with the operators for the three sites. However, permissions could not be obtained from the operators as the sites would still be in operation and the proposed SI works would have significant impact to their operations (e.g. the need to terminate/pause the current activities and demolition of existing structures).

8.7.3 As SI works at this EIA stage is not feasible, it is recommended that after the sites are handed over to the Project Proponent (PP), the PP's appointed consultant should revisit the sites to assess the latest site condition and to update the sampling and testing plan presented in the CAP. Supplementary CAP(s), reporting the latest site conditions, locations of hotspots and updated sampling and testing plan for the sites, should be submitted to EPD for approval at the later stage of the Project.

Potentially Contaminated Sites Inaccessible for Site Walkover

8.7.4 For the potentially contaminated sites inaccessible for site walkover (excluding the 90 warehouse sites), the actual sampling and testing plan could only be determined when the sites are available for detailed site walkover. However, the recommended minimum number of sampling locations for each of the concerned sites, taking into consideration the area of the potentially contaminated sites and adopting the regular grid sampling strategy in Section 2.4.1 of EPD's *Practice Guide*, has been summarised in Table 3.2 of the CAP given in **Appendix 8.1**. Extra sampling locations may be required after site walkovers are conducted, if additional potential sources of contamination (or 'hotspots') are identified within the concerned sites. The final sampling and testing plan, taking into account the abovementioned regular grid sampling strategy and any identified hotspots in the site walkover, will be reported in the supplementary CAP(s) at the later stage of the Project.

Table 8.5 Sampling and Testing Plan for the Potentially Contaminated Sites Accessible for Site Walkover

Site ID (Approx. Area)	Concerned Area (Approx. Area)	Grid Size (m)	Proposed Number of Boreholes	Rationale	Sampling Method	Sampling Matrix / Depth(1)		Testing Parameters				
								VOC(2)	SVOC(2)	Metals(2)	PCRs(2)	Other
A78 (Area: 1,860 m^2)	Maintenance Workshop (Area: 50 m^2)	6	**2** Borehole ID: A78 BH1 (refer to **Figure 8.6** for borehole location)	Although the whole area of A78 is 1,860 m^2, SI is only proposed to cover the maintenance workshop area which is approximately 50 m^2, hence only 2 boreholes are proposed.	Borehole to 2 m below the groundwater table or 6 m bgl	Soil	0.5 m bgl, 1.5 m bgl, 3.0 m bgl, and further at 3.0 m intervals upon encountering groundwater or 6 m bgl whichever is shallower, and above GW level if present.	Acetone, BTEX, MTBE and trichloroethene	PAHs	Chromium, copper lead, manganese, nickel, zinc	Full List	-
						GW	If present(3)	Acetone, BTEX, MTBE and trichloroethene	PAHs	-	Full List	
E63a (Area: 1,240 m^2)	Whole area of E63a (Area: 1,240 m^2)	13	**9** Borehole ID: E63a BH1 – BH9 (refer to **Figure 8.7** for borehole location)	According to the site representative, the site has been used as temporary storage of recycling materials for more than 20 years. A grid approach according to the Practice Guide is thus adopted for the SI works.	Borehole to 2 m below the groundwater table or 6 m bgl	Soil	0.5 m bgl, 1.5 m bgl, 3.0 m bgl, and further at 3.0 m intervals upon encountering groundwater or 6 m bgl whichever is shallower, and above GW level if present.	Full List	Full List	Full List	Full List	PCBs
						GW	If present(3)	Full List	Full List	Mercury	Full List	PCBs
E63b (Area: 5,140 m^2)	Petrol Filling Station (Area: 40 m^2)	6	**2** Borehole ID: E63b BH1 (refer to **Figure 8.8** for borehole location)	Although the whole area of E63b is 5,140 m^2, SI is only proposed to cover the petrol filling station which is approximately 40 m^2, hence only 2 boreholes are proposed.	Borehole to 2 m below the groundwater table or 6 m bgl	Soil	0.5 m bgl, 1.5 m bgl, 3.0 m bgl, and further at 3.0 m intervals upon encountering groundwater or 6 m bgl whichever is shallower, and above GW level if present.	BTEX and MTBE	PAHs	Lead	Full List	-
						GW	If present(3)	BTEX and MTBE	PAHs	-	Full List	-

Note:
(1) bgl = Below ground level; GW = groundwater
(2) Full list refers to the parameters as shown in Table 2.1 – RBRGs for Soil & Soil Saturation Limit and Table 2.2 – RBRGs for Groundwater and Solubility Limit under VOCs, SVOCs, Metals and PCRs in the Guidance Manual.
- BTEX includes benzene, toluene, ethylbenzene and total xylenes. MTBE - Methyl tert-butyl ether.
- PCBs - Polychlorinated biphenyls.

- PAHs include acenaphthene, acenaphthylene, anthracene, benzo(a)anthracene, benzo(a)pyrene, benzo(b)fluoranthene, benzo(g,h,i)perylene, benzo(k)fluoranthene, chrysene, dibenzo(a,h)anthracene, fluoranthene, fluorene, indeno(1,2,3-cd)pyrene, naphthalene, phenanthrene and pyrene.
Since RBRG value of Benzo(a)anthracene, Benzo(a)pyrene, Benzo(g,h,i)perylene, Benzo(k)fluoranthene, bis-(2-Ethylhexyl)phthalate, Dibenzo(a,h)anthracene, Indeno(1,2,3-cd)pyrene and Phenol were not available for groundwater, the captioned chemicals parameters would not be tested in groundwater sample.

(3) Samples will only be collected if groundwater is encountered during SI works.

(4) The recommended sampling and testing strategy is only tentative and is subject to review after land resumption and site re-appraisal.

8.8 Evaluation of Land Contamination Impacts

8.8.1 Based on the site appraisal, 480 potentially contaminated sites have been identified. However, the land contamination issues within these concerned sites are considered not insurmountable due to the following factors:

Size of individual sites and related operation scale are relatively small and the extent of any potential contamination is anticipated to be localised rather than widespread

8.8.2 Based on the site appraisal, more than half of the potentially contaminated sites are currently used as open area storage, container storage and warehouse. As discussed above, warehouses may not be contaminated if they only stored non-contaminating goods such as furniture or toys. For container storage and open area storage, a large portion of their areas are typically used for goods storage with possibly smaller portion for potentially contaminating activities such as vehicle / equipment maintenance area and the associated chemical handling/storage. If the main types of goods stored on-site are not potential sources of contamination, land contamination, if indeed present, for these land types would likely be localised.

8.8.3 In addition, the potentially contaminated sites are not large scale polluting installations / facilities such as oil depots and power plants. Apart from the sites which are used as open area storage, container storage and warehouse, more than 70% of the remaining sites are less than 3,000 m^2 in area which is considered relatively small in scale. Therefore, the contamination extent, if any, caused by the operations of the identified potentially contaminated sites is anticipated to be localised.

Chemicals of Concern identified are readily treatable using established physical, chemical and biological techniques

8.8.4 Based on the site appraisal, the potential COCs include metals, VOCs, SVOCs, PCRs and PCBs. Remediation methods that have demonstrated to effectively treat the aforementioned soil and groundwater contaminants include biological treatment and physical / chemical treatment. Details of the treatment methods are provided in Section 5 of the CAP given in **Appendix 8.1**.

Local remediation experience is sufficient to deal with the nature of the possible contaminants

8.8.5 Soil contaminated with the abovementioned COCs had successfully been remediated in Hong Kong using proven remediation techniques. Notable local remediation projects include the followings:

- Decommissioning of Kai Tak Airport North Apron;
- Decontamination works at the Cheoy Lee Shipyard;
- Reclamation works at North Tsing Yi Shipyard site;
- Decommissioning of Kwai Chung Incinerator; and
- Isolated sites in the Deep Bay Link project.

Further Works

8.8.6 Further works, including site re-appraisal, SI works and submission of supplementary CAP(s), CAR(s) / RAP(s) and RR(s), are recommended to identify the presence, nature and extent of contamination and, if necessary, to remediate the contaminated soil and groundwater. Any soil/groundwater contamination would be identified and properly treated prior to the development of the concerned sites. Details of the further works are discussed in **Section 8.9**.

8.9 Recommendation of Further Works

8.9.1 The further works should be carried out in accordance with Guidance Note, Practice Guide and Guidance Manual (refer to **Section 8.2**).

Identified Potentially Contaminated Sites

8.9.2 The identified potentially contaminated sites are still in operation and as the development will only commence in stages from 2019 to 2037/38, there might be change in land use prior to the development. Furthermore, the majority of these sites were inaccessible at the time of reporting to assess the site conditions / hotspots, formulate the detailed sampling and testing plan or conducting the actual SI works.

8.9.3 Therefore, prior to development of these sites, the PP should appoint a consultant to re-appraise these sites to ascertain the initial evaluation and recommendation reported and update the corresponding findings (e.g. locations of hotspots) and sampling and testing requirements presented in this CAP. For the 90 'warehouse' sites discussed in **Section 8.6.3**, the site re-appraisal should also confirm the necessity for SI works and, if required, formulate the sampling and testing strategy. Supplementary CAP(s), incorporating the findings of the site re-appraisal and the updated sampling and testing strategy, should be prepared and submitted to EPD for approval prior to conducting any SI works.

8.9.4 SI works should then be carried out according to the supplementary CAP(s). CAR(s) and, if contaminated soil and/or groundwater identified, RAP(s) should be prepared and submitted to EPD for approval. Any contaminated soil and groundwater should be treated according to EPD's approved RAP(s) and RR(s) should be submitted to EPD for agreement after completion of the remediation works. Regular site audit during construction phase is required to ensure the proposed mitigation measures as recommended in the future RAP(s) are implemented effectively. No development works shall be commenced prior to EPD's agreement of the RR.

Remaining Non-Contaminated Sites

8.9.5 Similarly, as the development will only commence in stages from 2019 to 2037/38, it is possible that the land uses of the non-contaminated sites could change to contaminated land types. Hence, after the sites are handed over to the PP for development, the PP's appointed consultant should also revisit these sites to assess the latest land uses and site conditions as well as to ascertain the initial evaluation and recommendation reported in the CAP. If any of these sites are found to have potential land contamination issues, the PP's appointed consultant should also prepare and submit supplementary CAP(s) to EPD for approval prior to conducting any SI works.

8.9.6 SI works should then be carried out according to the supplementary CAP(s). CAR(s) and, if contaminated soil and/or groundwater identified, RAP(s) should be prepared and submitted to EPD for approval. Any contaminated soil and groundwater should be treated according to EPD's approved RAP(s) and RR(s) should be submitted to EPD for agreement after completion of the remediation works. Regular site audit during construction phase is required to ensure the proposed mitigation measures as recommended in the future RAP(s) are implemented effectively. No development works shall be commenced prior to EPD's agreement of the RR.

8.10 Evaluation of Residual Impacts

8.10.1 Referring to **Section 8.8**, the land contamination issues associated with the potentially contaminated sites are considered surmountable if the recommended further works in **Section 8.9** were followed and contaminated soil and groundwater (if any) were properly treated using appropriate remediation methods and according to EPD's approved RAP. However, any localised soil/groundwater contamination within these sites, if left untreated, could potentially pose health risks to existing or future occupants.

8.10.2 By implementing the recommended further works under this Project, the actual contaminated site(s) within the assessment area would be located and any contaminated soil and groundwater would be identified and treated. The recommended further works would not only minimise the health risks to the future occupants arising from the exposure of the contaminated soil and/or groundwater, it would also provide the opportunity to treat the contaminated soil / groundwater using proven remediation techniques for reuse as useful materials (such as backfilling materials); thereby minimising the amount of waste disposing into the already depleting landfills in Hong Kong and achieving a more sustainable development.

8.10.3 Furthermore, the Project would allow the conversion of the contaminated site(s) into land that are safe for more optimal development (e.g. residential development). The more optimal development would assist in addressing Hong Kong's long-term housing demand and other land use needs.

8.11 Environmental Acceptability of Schedule 2 Designated Projects

8.11.1 This Project (i.e. HSK NDA) is a Schedule 3 Designed Project (DP) under the EIAO, whilst there will be 12 Schedule 2 DPs. The details of these Schedule 2 DPs are presented in **Section 1.4 (**i.e. **Table 1.1)** and **Section 2.4**, and the locations of these Schedule 2 DPs are shown in **Figure 1.2**.

8.11.2 The identified potentially contaminated sites are grouped under each Schedule 2 DP and the findings are summarised in **Appendix 8.2**. As the land contamination potential of the sites listed in **Appendix 8.2** have been assessed under this EIA Study, the land contamination issue within these sites are considered surmountable (refer to **Section 8.8** for details). Prior to the commencement of construction of each Schedule 2 DP, the recommended further works under this EIA Study (including site re-appraisal, SI works and if required, remediation works (refer to **Section 8.9** for details)) will be carried out for the concerned sites listed in **Appendix 8.2**. Any soil / groundwater contamination would be identified and properly treated prior to the development of the concerned sites. No residual land contamination impacts are anticipated, it is considered environmentally acceptable if the recommended further works in **Section 8.9** were followed.

8.12 Conclusion

8.12.1 This land contamination assessment examined the potential contaminative land use within the assessment area and their potential impacts to future land use.

8.12.2 Based on the findings from desktop review (e.g. review of historical aerial photos and relevant information from EPD and FSD), helicopter reconnaissance and site surveys, 480 potentially contaminated sites have been identified. Findings of the desktop review and site surveys as well as the proposed soil and groundwater sampling and testing strategy are presented in the CAP enclosed in **Appendix 8.1**.

8.12.3 The majority of the potentially contaminated sites were inaccessible for site walkover at the time of reporting to assess the site conditions. For sites that are accessible for site walkover, permission could not be obtained from the site operators to carry out the SI works. Furthermore, as the identified potentially contaminated sites are still in operation and the development will only commence in phases from 2019 to 2037/38, there may be change in land use prior to development within both the potentially contaminated and non-contaminated sites. Further works, including site re-appraisal, SI works as well as submission of supplementary CAP(s), CAR(s) and RAP(s) for EPD's approval are recommended to be carried out after the sites are handed over to PP for development. If contaminated soil and/or groundwater were identified, remediation should be carried out according to EPD approved RAP(s) and RR(s) should be submitted to EPD for agreement after completion of the remediation works. No development works shall commence prior to EPD agreement of the RR. The further works would need to follow EPD's Guidance Manual, Guidance Note and Practice Guide and according to **Section 8.9**.

8.12.4 Based on the findings from the assessment, 253 potentially contaminated sites are currently used as open area storage, container storage and warehouse. Warehouse sites may not be contaminated if they are used to store general household goods (e.g. furniture and toys). Container storage and open area storage, on the other hand, typically comprise a large portion of area for goods / container storage with possibly smaller portion for potentially contaminating activities such as vehicle / equipment maintenance area and the associated chemical handling/storage. The contamination (if any) is therefore expected to be localised if the main types of goods stored on-site are not potential sources of contamination. In addition, the land uses of the remaining identified potentially contaminated sites are not large scale polluting installations / facilities, such as oil depot and power plant, which further support that the contamination (if any) would be localised rather than widespread.

8.12.5 The COCs that have been identified with the potential to be present at the potentially contaminated sites include metals, VOCs, SVOCs, PCRs and PCBs. These COCs are readily treatable using established physical, chemical and biological techniques and soil contaminated with the abovementioned COCs had successfully been remediated in Hong Kong using proven remediation techniques. By implementing the recommended further works, the actual contaminated site(s) within the assessment area would be located and any contaminated soil and groundwater would be identified and treated.

8.12.6 Given the above, land contamination impacts are therefore considered not insurmountable to future occupants.

8.12.7 The implementation of the recommended further works under this Project would clean up any contaminated site(s) identified within the assessment area. The recommended further works would not only minimise the health risks to the future occupants arising from the exposure of the contaminated soil and/or groundwater, it would also provide the opportunity to treat the contaminated soil / groundwater using proven remediation techniques for reuse as useful materials (such as backfilling materials); thereby minimising the amount of waste disposing into the already depleting landfills in Hong Kong and achieving a more sustainable development.

8.12.8 Furthermore, the Project would allow the conversion of the contaminated site(s) into land that are safe for more optimal development (e.g. residential development). The more optimal development would assist in addressing Hong Kong's long-term housing demand and other land use needs.